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Outcomes of general anesthesia for noncardiac surgery in a series of patients with Fontan palliation

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Summary

Objectives: To describe the experience of a single, tertiary care institution in the care of patients with Fontan physiology undergoing anesthesia for noncardiac surgery.
Background: The Fontan procedure was developed in 1971 to palliate patients with univentricular cardiac physiology leading to long-term survival of these patients, who may now present as adults for noncardiac surgery.
Methods: We retrospectively reviewed the medical records of Fontan patients 16 years and older who underwent general anesthesia for noncardiac surgery at Mayo Clinic in Rochester, Minnesota. Preoperative data, perioperative course, intraoperative and postoperative hemodynamic, pulmonary, cardiovascular, and renal complications were described.
Results: Thirty-nine general anesthetics were administered to 31 patients for noncardiac surgery after Fontan palliation. Perioperative complications occurred in 12 of the 39 (31%) noncardiac surgeries, and there was one postoperative death that occurred on day 13 after ventral hernia repair. The two patients who had complications that did not resolve (long-term dialysis and death) had ejection fractions well below the mean for the group (22% and 28%).
Conclusion: It may be more appropriate for Fontan patients to undergo anesthesia for noncardiac surgery in a tertiary institution, particularly patients with an ejection fraction of <30%. Intraoperative arterial blood pressure monitoring and overnight admission are likely appropriate for most cases.

Introduction

The Fontan procedure was developed in 1971 to offer surgical palliation to patients with univentricular cardiac physiology (1). Globally, patients have 60–93% chance of being alive at 10 years after Fontan palliation (2–4) and up to 87% of patients will be alive 20 years after operation (5).

Given the above, a number of patients with Fontan anatomy and physiology may present for noncardiac surgery. The purpose of this study is to describe the experience of a single, tertiary care institution in the care of patients with Fontan physiology undergoing anesthesia for noncardiac surgery.

Methods

Following Institutional Review Board approval, we retrospectively examined the medical records of Fontan patients who underwent general anesthesia for noncardiac surgery at Mayo Clinic in Rochester, MN, USA. One thousand one hundred and thirty-three patients who

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underwent prior Fontan surgery at Mayo Clinic or at an outside institution were identified from the Mayo Clinic Cardiothoracic Surgery database (1972–2009) as well as from the Mayo Clinic Congenital Heart Disease Clinic database (1988–2009). Noncardiac surgeries performed in this cohort between 1988 and 2009 were obtained from the Anesthesia database and the Surgical Information Recording System. Patients included in this study previously underwent Fontan palliation and were at least 16 years of age at the time of the noncardiac surgery. Noncardiac surgeries were excluded if the patient underwent a takedown of the Fontan or a cardiac transplant prior to the noncardiac surgery. Diagnostic procedures such as endoscopy or bronchoscopy and procedures without an incision such as closed fracture reduction were excluded. Procedures performed during the same admission as the Fontan surgery as well as procedures in patients already intubated in the ICU for a prolonged period leading up to the noncardiac surgery were also excluded. One noncardiac surgery in a Fontan patient in this series has previously been described in a letter to the editor (6).

Preoperative information collected included gender, cardiac morphology, date of Fontan surgery, type of Fontan palliation, American Society of Anesthesiology (ASA) physical status at the time of surgery, ejection fraction of systemic ventricle within a year of noncardiac surgery, hemoglobin, platelet count and creatinine within a month of noncardiac surgery, preoperative medications, and preoperative oxygen saturation. Surgical data collected included date of noncardiac surgery, surgical procedure, emergent or elective nature of surgery, whether laparoscopic or open technique was used where applicable and occurrence of bleeding complications requiring reoperation. Anesthetic data collected included hemodynamic monitoring and airway management techniques, transfusion of blood products, antifibrinolytic use, use of antiarrhythmics, vasoconstrictor or inotropic support, and pulmonary vasodilators use as well as intraoperative complications. Postoperative data collected included postoperative disposition, use of antiarrhythmics, vasoconstrictor or inotropic support and pulmonary vasodilators, postoperative complications, postoperative ventilation requirement and duration, intensive care unit length of stay, hospital length of stay, and readmission after discharge. Postoperative disposition to intensive care and to intermediate care was defined as admissions to monitored care. Complications were defined as death, cardiopulmonary resuscitation, dysrhythmias, need for electrical cardioversion, new ST segment depression, new ST segment elevation, acute heart failure, sustained hypotension, postoperative reintubation, acute pulmonary edema, pulmonary embolism, hemorrhagic stroke, embolic stroke, acute renal failure and use of an antiarrhythmic, vasoconstrictor or inotropic infusion or pulmonary vasodilators if this was not performed prophylactically. Study data were collected (JAR, CBG and DWB) and managed using REDCap electronic data capture tools hosted at Mayo Clinic (7).

Data were represented as median (range) for continuous variables and as counts for categorical variables. The distributions of all variables were examined, and given the skewed distributions, the data were analyzed using nonparametric methods to decrease the influence of outliers. *Post hoc* analysis was performed investigating for association between preoperative ejection fraction and occurrence of complications, monitored care length of stay and hospital length of stay. The rank sum test was used to test for differences in pre-ejection fraction between patients who experienced complications and those who did not. Spearman rank correlation was used to test for association between ejection fraction and monitored care length of stay and hospital length of stay. P-values < 0.05 were considered statistically significant.

Results

A total of 39 general anesthetics were administered to 31 patients (18 men, 13 women) for noncardiac surgery at Mayo Clinic following Fontan palliation. Twenty-four patients had one noncardiac surgery, six patients had two noncardiac surgeries, and one patient had three noncardiac surgeries. Median age at the time of noncardiac surgery was 26 years (range, 16–57 years). Twenty-six of the 31 patients underwent Fontan palliation at Mayo Clinic. The median interval between Fontan operation and subsequent noncardiac surgery was 15 years (range, 2–25 years). Preoperative data including medications and test values are presented in Table 1.

Patients underwent a variety of noncardiac surgeries, with four of them being emergent in nature (Table 2). Three procedures were performed laparoscopically. A fourth was performed laparoscopically and converted to open technique because of technical reasons. The median anesthetic duration was 138 min (range, 48–612 min). In 20 of the 39 procedures, patients were managed with noninvasive monitors only. In the other 19 procedures, 11 patients had arterial catheters placed, 7 had arterial catheters with central venous access, and one patient received an arterial catheter, central venous catheter and transesophageal echocardiography (Table 2).

Four patients underwent posterior thoracolumbar fusion for scoliosis. Estimated blood loss during these

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Table 1 Demographic and preoperative data in 31 Fontan patients undergoing 39 noncardiac surgeries

Age at surgery, years	
Age at Fontan surgery	9 (2–40)
Age at noncardiac surgery	25 (16–57)
Years from Fontan to noncardiac surgery	14 (2–25)
Gender	
Female	13
Male	18
Ventricular morphology (systemic ventricle)	
LV	20
RV	11
Type of Fontan repair	
RA to PA connection	19
Intra-atrial conduit	9
Extracardiac conduit	3
ASA physical status prior to NCS	
2	5
3	28
4	5
Not documented	1
Preoperative test results	
Preoperative oxygen saturation (%)	94 (82–100)
Hemoglobin ($g \cdot dl^{-1}$)	15.4 (9.3–20.7)
Platelets ($\times 10^9 \cdot l^{-1}$)	157 (89–535)
Creatinine ($mg \cdot dl^{-1}$)	0.9 (0.7–2.3)
Systemic ventricle EF (%)	50 (20–64)
Medications	
Antiarrhythmics	28
Diuretics	20
Anticoagulants	23
Aspirin	4
Coumadin	18
Dipyridamole	1
ACE inhibitors	25
Beta-blockers	5

Data given as count for categorical data and median (range) for continuous data. LV, left ventricle; RV, right ventricle; RA, right atrium; PA, pulmonary artery; ASA, American Society of Anesthesiology; NCS, noncardiac surgery; EF, ejection fraction; ACE, angiotensin convertin enzyme.

Table 2 Perioperative data for 39 noncardiac surgeries in 31 Fontan patients

Type of surgery	
Vascular	2
Digestive system	7
Nose/mouth/pharynx	3
Musculoskeletal	4
Nervous system	2
Hernia/Lymphatic	5
Urinary system	1
Male genitalia	5
Female genitalia	7
Integumentary	2
C-section	1
Emergency surgery	4
Airway type	
ET tube	35
Laryngeal mask airway	3
Mask	1
Monitors	
ASA	20
ASA, arterial line	11
ASA, arterial line, CVP	7
ASA, arterial line, CVP, TEE	1
Postoperative disposition	
Outpatient	11
Inpatient floor care	8
Inpatient monitored care unit	20
Length of stay (days)	
Duration of monitored care	2 (1–30)
Duration of total hospital stay	2 (0–35)
Length of Stay	
0 days	11
1 day	6
2 or more days	22

Data given as count for categorical data and median (range) for continuous data. ASA monitors included non-invasive blood pressure, electrocardiogram, pulse oximetry, and capnography. ET, endotracheal; ASA, American Society of Anesthesiologists standard monitors; CVP, central venous line; TEE, transesophageal echocardiogram.

four surgeries was 1350, 1500, 4500, and 8000 ml. In 28% of the procedures, the patients were treated as outpatients. None of the outpatients were readmitted to the hospital within 72 h of discharge. After 51% of procedures, the patient was admitted to monitored care postoperatively (Table 2). Three of these patients required endotracheal intubation for <24 h postoperatively.

Perioperative complications

There were perioperative complications in 12 of the 39 (31%) noncardiac surgeries. These occurred in 11 of the 31 patients (Table 3). Median preoperative test results for the 12 noncardiac surgeries where complications occurred were ejection fraction 45%, creatinine 0.9 $mg \cdot dl^{-1}$, hemoglobin 15.7 $g \cdot dl^{-1}$, and oxygen saturation 93%. Perioperative complications occurred intraoperatively in four cases, postoperatively in six cases, and both intraoperatively and postoperatively in two cases. The majority of complications were hemodynamic disturbance or fluid overload, and there were two cases of postoperative arrhythmia requiring treatment. One patient was reintubated in the recovery room and was extubated later that day. No patients had bleeding complications requiring reoperation.

Three patients developed acute renal failure of which one recovered without intervention. The second required long-term dialysis after a partial nephrectomy for renal cell carcinoma and was intubated for 16 days but ultimately was discharged to home on postoperative day 35.

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Table 3 Perioperative details of 11 Fontan patients with complications during 12 noncardiac surgeries

	Cardiac history			Preoperative data				Intraoperative data			Postoperative data				
Patient Surgical procedure (age in years at noncardiac surgery)	Single vent & Fontan type	Age at Fontan (Y)	Time s/p Fontan (Y)	EF (%)	Hb (g.dl)	Cr (mg·dl^{-1})	SpO_2 (%)	Monitors	Blood loss (ml)	Prophylactic infusions	Disposition	ICU LOS [ventilation days]	Hospital LOS (days)	Complications	Treatment
Patient 1 Open cholecystectomy, umbilical herniorrhaphy (50 y)	LV RA–PA	30	20	53	9.3	1.1	95	ASA, Art, CVP	650	Dobutamine	ICU	6	9	Intraoperative hypotension; postoperative fluid overload	Phenylephrine infusion; Diuresis, oxygen
Patient 2 Varicose vein stripping (39 y)	LV RA–PA	25	13	N/A	12	0.9	83	ASA, Art	N/A		ICU	2	6	Postoperative fluid overload	Diuresis, oxygen
Patient 3 Craniotomy, aneurysm clipping (23 y)	LV RA–PA	9	14	60	14.5	0.7	N/A	ASA, Art, CVP, TEE	N/A	Dopamine, nitroprusside	ICU	5	11	Postoperative hypotension	Dopamine infusion (4 days)
Patient 4 Craniotomy, resection orbital osteoma (24 y)	LV RA–PA	4	20	53	17	0.7	97	ASA, Art	400		ICU	2 [vent 1 day]	6	Respiratory distress	Reintubation (recovery room)
Patient 5 Total abdominal hysterectomy (24 y)	LV Intra-atrial	5	19	33	16.5	0.7	87	ASA, Art, CVP	100		ICU	2	13	Acute renal failure (day 4)	Resolved
Bilateral salpingo-oopherectomy (26 y)	LV Intra-atrial	5	21	43	15.8	0.7	85	ASA, Art	600		ICU	1	22	Hypotension (recovery room)	Fluid, ephedrine
Patient 6 Excision retroperitoneal paraganglioma (25 y)	RV RA–PA	8	16	45	16.6	0.9	N/A	ASA, Art	600	Milrinone, inhaled epoprostenol	ICU	1	7	Undiagnosed pheo	Phenylephrine, nitroprusside
Patient 7 Posterior thoracolumbar fusion (18 y)	RV Intra-atrial	5	13	50	13.8	0.7	92	ASA, Art, CVP	1500		ICU	2	6	Intraoperative hemodynamic instability	Milrinone infusion
Patient 8 Emergency biopsies thigh, cellulitis (33 y)	LV RA–PA	20	13	45	16.2	0.9	94	ASA	N/A		ICU	2	17	Intraoperative hypotension	Dopamine infusion
Patient 9 Posterior thoracolumbar fusion (17 y)	RV Extracardiac	5	11	20	15.7	1.3	91	ASA, Art, CVP	1350	Milrinone	ICU	8	8	Suspected ventricular failure	Inhaled epo, milrinone inf (intraop-day 7)

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Table 3 (Continued)

Patient Surgical procedure (age in years at noncardiac surgery)	Cardiac history			Preoperative data				Intraoperative data			Postoperative data				
	Single vent & Fontan type	Age at Fontan (Y)	Time s/p Fontan (Y)	EF (%)	Hb (g.dl)	Cr (mg·dl^{-1})	SpO_2 (%)	Monitors	Blood loss (ml)	Prophylactic infusions	Disposition	ICU LOS [ventilation days]	Hospital LOS (days)	Complications	Treatment
Patient 10 Partial nephrectomy (57 y)	LV RA–PA	34	23	22	13.7	1.4	100	ASA, Art, CVP	700	Milrinone	ICU	30 [vent 1 + 16 days]	35	Prolonged intraoperative hypotension; postoperative arrythmias; respiratory failure (day 6); ARF (day 6)	Epinephrine, dopamine inf (1 day); amiodarone infusion; reintubation, inhaled epo; CRRT
Patient 11 Ventral hernia repair (43 y)	LV RA–PA	30	13	28	15.7	1.0	95	ASA	N/A		ICU	13 [vent 1 day]	13	Postoperative hypotension, arrhythmia (VT), seizure, acute heart failure (day 1), pneumonia (day 5), sepsis, multiorgan failure (day 7), death (day 13)	PACU: art line, epinephrine, dopamine, dobutamine infusion (1 day), intubation (1 day) ICU: NIPPV, antibiotics (day 5), CRRT (day 7), pressors, CPR (day 13)

Single vent, single ventricle; y, years; LV, left single ventricle; RV, right single ventricle; RA–PA, right atrium to pulmonary artery connection; N/A, not available; ASA, American Society of Anesthesiology standard monitors; art, arterial line, CVP, central venous line; TEE, transesophageal echocardiogram; ICU, intensive care unit; LOS, length of stay; vent, duration of ventilation, pheo, pheochromocytoma; ARF, acute renal failure; epo, epoprostenol; inf, infusion; intraop, intraoperative; CRRT, continuous renal replacement therapy; NIPPV, noninvasive positive pressure ventilation.

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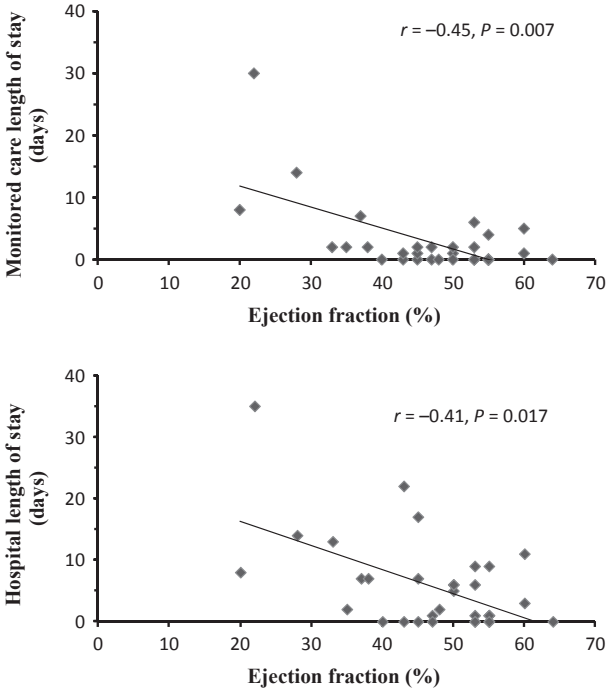


Figure 1 Pre-operative ejection fraction versus monitored care length of stay and hospital length of stay.

The third patient developed multiorgan failure and died on postoperative day 13 after hernia surgery.

Postoperative death occurred in one Fontan patient after an elective ventral hernia repair with mesh for intractable pain. This patient had a preoperative ejection fraction of 28% and had persistent atrial arrhythmias controlled on amiodarone. Intraoperative course was stable; however, upon arrival to PACU, he became hypotensive, developed intermittent episodes of nonsustained ventricular tachycardia and had a transient grand mal seizure. The patient was reintubated, an arterial line placed, and epinephrine, dopamine, and dobutamine infusions initiated. Echocardiogram revealed a severely dilated right atrium with compression of the right upper pulmonary veins. The patient was weaned from inotropic support and extubated within 24 h. Over the next several days, he developed pneumonia with worsening respiratory status, renal failure requiring continuous veno-venous hemodiafiltration, hypotension refractory to multiple vasopressors, and hepatitis. He died on day 13, and autopsy confirmed acute heart failure, acute respiratory failure with Pseudomonas aeruginosa pneumonia, acute renal tubular necrosis, acute massive hepatic necrosis (consistent with diffuse hypoperfusion injury), and sepsis.

Preoperative ejection fraction was significantly lower in those who experienced complications as compared to those who did not (median, 45% vs 53%; $P = 0.045$). Preoperative ejection fraction was negatively correlated with both monitored care length of stay (Spearman rank correlation $r = -0.45$, $P = 0.007$) and hospital length of stay ($r = -0.41$, $P = 0.017$; Figure 1).

Discussion

To our knowledge, we have described the largest single series of patients with Fontan physiology undergoing noncardiac surgery. Complications occurred in 31% of surgeries, and there was one postoperative death after ventral hernia surgery for intractable pain. Preoperative ejection fraction predicted occurrence of complications, monitored care length of stay, and hospital length of stay. In addition, the two patients who had complications that did not resolve (long-term dialysis and death) had ejection fractions well below the median for the group (22% and 28%) and both had chronic atrial arrhythmias. Fontan patients, particularly those with depressed ventricular function and a history of atrial arrhythmias, may have very tenuous hemodynamics and appear to be vulnerable to hemodynamic instability even with relatively benign noncardiac surgery. Similar types and frequencies of complications occurred in patients with left versus right ventricular morphology and with different types of Fontan palliation.

Patients with Fontan physiology undergoing noncardiac surgery present a number of potential concerns for the perioperative team. Perhaps most important is the dependence on venous pressure to provide pulmonary perfusion and ventricular filling. This anatomy is critically dependent on the maintenance of adequate preload, low pulmonary vascular resistance, competency of the atrioventricular valve, and adequate contractility of the systemic ventricle to maintain cardiac output. Fontan patients are at risk for arrhythmias, and maintenance of sinus rhythm is very important for those with tenuous hemodynamics, especially those with a depressed ejection fraction. Of additional concern is complex coagulation abnormalities frequently encountered in these patients (8,9) as well as altered liver and renal function and autonomic nervous system dysfunction (10–12).

Fontan patients have complex coagulation issues and are predisposed to both hemorrhage and thrombosis. The increased risk for bleeding is attributed to decreased hepatic synthesis of coagulation factors, particularly factor VII (8) because of decreased hepatic function and protein losing enteropathy. These patients are also frequently on chronic anticoagulation. Increased risk of thrombosis is attributed to sluggish venous flow and polycythemia as well as increased platelet reactivity (9) and elevated factor VIII (8). This raises concern that

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these patients may be at increased risk of perioperative bleeding complications and thrombotic events, including deep venous thrombosis with pulmonary embolism, myocardial infarction, and cerebrovascular events. In the present case series of 38 noncardiac surgeries, there were no cases of perioperative thrombotic events.

One case report describing scoliosis surgery in patients who had a prior Fontan procedure highlighted the intraoperative complication of massive venous bleeding, aggravated by positive pressure ventilation and prone positioning (13). Four patients in our case series underwent scoliosis surgery with massive blood loss occurring in two of these patients.

Two case reports of patients with single ventricle physiology undergoing uncomplicated laparoscopic procedures discuss the theoretical detrimental effects of pneumoperitoneum, reverse Trendelenburg positioning and carbon dioxide absorption on venous return and the maintenance of cardiac output in these patients (14,15). Complications of laparoscopy including CO_2 embolism or pneumothorax would be significantly worse in patients whose pulmonary venous return is passive. A fenestration between the Fontan conduit and common atrium may add the risk of paradoxical, systemic embolic phenomena. In our series of cases, three patients underwent laparoscopic surgery and a fourth had initial laparoscopy performed, without complication.

Two case reports of adrenalectomy for pheochromocytoma in a Fontan patient described postoperative complications in this patient including refractory atrial arrhythmias required multiple cardioversions and ascites requiring diuresis and fluid restriction (6,16). One patient in our case series underwent resection of a pancreatic mass, which was discovered to be an undiagnosed paraganglioma. The management of this case has been previously reported (6).

These noncardiac surgeries described in Fontan patients occurred during a 22-year period at a tertiary referral center and were obtained from a database of 1133 Fontan patients. The frequency of noncardiac surgery in Fontan patients cannot be ascertained from this case series as many of the patients may have traveled to our institution for their Fontan surgery and subsequently may have had noncardiac surgeries elsewhere. However, noncardiac surgeries in Fontan patients at Mayo Clinic do appear to be relatively infrequent. The type of noncardiac surgeries, which these patients underwent, may not be representative as some of the Fontan patients may have undergone operations closer to home.

In this tertiary care setting, the majority of patients were managed with standard ASA monitors only and many patients underwent ambulatory anesthesia and surgery. It is unclear whether the patient who died may have benefited from having more invasive intraoperative monitors (e.g., invasive blood pressure monitoring) enabling earlier recognition of deterioration. Not all complications were evident on the day of surgery; however, none of the outpatients were readmitted postoperatively. The current series is too small to draw conclusions on the appropriateness of same day surgery in patients with Fontan physiology.

Limitations

The limitations of this study include those inherent to a retrospective study. For example, it was not possible to distinguish levels of monitored care from the medical record. Thus, admissions to the intensive care and to the progressive care units were combined as admissions to monitored care units. It was also not possible to determine whether monitored care admissions were planned preoperatively or if they were unplanned. The limited number and variation of procedures and patient presentations allow only for descriptive statistics and do not allow for the calculation of incidence rates. One therefore cannot conclude based on the absence of major complications in this series that noncardiac surgery in this patient populations can be performed in any setting. The observations of this series of cases at a tertiary referral hospital cannot be extrapolated to other settings.

Conclusion

This description of perioperative care of the adult patient with Fontan physiology suggests that it may be more appropriate for these patients to undergo anesthesia for noncardiac surgery in a tertiary institution with congenital heart disease specialists and under the care of an anesthesiologist familiar with Fontan hemodynamics. This is especially true for those patients with an ejection fraction of <30% who had longer monitored care stays, long-term complications, and death. The risks and benefits of undergoing noncardiac surgery in this setting must be considered carefully. The necessary degree of intraoperative and postoperative monitoring is unclear but intraoperative arterial blood pressure monitoring and overnight admission are likely appropriate for most cases.

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Conflict of interest

No conflicts of interest declared.

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